

Methodological Continuity Plan — Part 2: the Parameterized Computational Anticipation (ACP) Method, from Simulated Data to Derived Research

Open Prion & Molecular Engineering Consortium*

Abstract

Part 1 demonstrated the design thesis (V127 platform, $\theta^*=0,333$, reproduced in two environments). Part 2 — this manifest — is the realized thesis and formalizes the research method: ACP, Parameterized Computational Anticipation, steps P0–P6 (identification → parametrization → execution → harvest → locked prognosis → immediate derived research → optional confrontation). Mandatory axis: everything here is simulation and is declared as such; we do not promise to apply or validate — if real data ever prove analogous, the subsequent steps are already taken (anticipation banked). Validity Basis is mandatory (BLOCKED check): simulation does not replace the laboratory — essential for real absorption — and every datum has full lineage (who → species → cross-validation → code → parametrization → result). The thesis results are the executed [SIM] findings, including the calibrated θ_{obs} estimator and the finding that biomass ratio carries the information the saturated radius loses. [NO YEAR]: simulation prognoses operate in future time by nature. Method, not selection: no laboratory is chosen.

1 The named method: ACP

Today it is possible to continue research by simulation: validated published data as input, auditable deterministic execution, prognoses locked before any measurement — without replacing or waiting for the laboratory. ACP is a sibling of meta-analysis, physics-based modeling and in-silico trials, and is distinguished by locking the prognosis before measurement and labeling simulation in every output (tiers).

2 Validity Basis (mandatory)

Triad: (i) simulation-based thesis does not replace laboratory validation — the laboratory is essential for real absorption; (ii) continuation comes from prognoses: future laboratory exams equivalent to the simulated mean

data and information already anticipated and immediately applicable; (iii) experimental rigor is converted to the computational environment with complete data lineage per datum (six-row table in the PT master).

3 The thesis in experimental form (analogical mapping)

A thesis equal in form to the experimental one: instead of people, the research subject is the set papers + sources + code + simulation, documented identically. Participants → published sources (38 E-registered); recruitment → audited SLR; consent → open provenance; calibrated instruments → self-tested code (100%/0.5%); protocol → P1–P2; collected events → archived runs [SIM] (2-environment reproduction); records → thesis registry (54 claims · 48 numeric facts · lineage); pre-specified analysis → P3 + calibrated estimator; implications → P4 + P5 + P6; audit → guardian-2 across four dimensions. The produced datum remains [SIM] — the role changes, the label does not.

4 The thesis results are the results of the simulation already performed

θ^* ; design rules; sensitivity harvest with discriminating prediction (C_{50} insensitive $10\times$; functional form falsifiable by dose-response); two-lens probabilistic frame (registry-bound prior); calibrated θ_{obs} (pre-declared verdict; decision-boundary bias -0.008 ; v1.1 honestly rejected); two-environment reproduction; and the derived decision table — radius margin saturates early (70.2% at $\kappa=1.5$), biomass ratio is the informative coordinate ($48\rightarrow1.25$).

5 Components M1–M5 (reframed)

M1 estimator — the continuity instrument (executed, passed). R1 results-as-results — [SIM] findings are the thesis results. M3 re-parametrization loop — recalibrates only what the data informs; compares, never re-trains. M2 freeze/GATE-F — optional dormant extension. M4 partner selection — SLR-analogue method for replicability (non-decisional pilot that corrected the

*Corresponding: Camilla N. (CRedit in-repo). AI under continuous human supervision; not an author (SW-S01/03). Companion to Part 1 (release v3.0), unmodified. Gated: guardian.py –profile part2 (R3-BASE-VALIDADE BLOCKED). PT is the master.

Table 1: ACP — formal steps P0–P6.

Step	Name	Input	Procedure	Output
P0	identification	focal question	audited SLR of validated data	E-registered base
P1	parametrization	base	derivation w/ provenance	parameterized models
P2	execution	models	deterministic engines, self-tests	runs [SIM]
P3	harvest	runs	pre-declared criteria	results [SIM]
P4	prognosis	results	predictions locked by release	thresholds/decision-table
P5	derived research	reasonable values	new questions proceed immediately	[SIM] agenda
P6	confrontation (optional)	analogous real data, if any	comparison to anchor (never retraining)	subsequent steps already adva

prior). M5 infrastructure — guardian with surface profiles and decalogue.

6 Promises and limits

Promises: ACP as a replicable method, partly demonstrated, to continue research today. Declares: results are simulation and remain labeled; real validation not promised here; analogous future data mean subsequent steps already advanced (P6). Limits: single-rater structured judgments; current-grid calibration; tool divergences documented in AUDIT_NOTES. Express validation by the author — recorded (28/08, PR #2 merge commit).

Artifacts inventory (verifiable on disk)

scripts: part2_theta_obs_{v1,pooled,v11}.py ·
results: part2_results/ · method-docs: freeze
checklist (dormant), REPARAM_LOOP, PART-
NER_SELECTION_PROTOCOL v2.1 · gate:
guardian.py --profile part2 · shared registry: 54
claims · 38 sources · 48 numeric facts · AST executable
6/6.